

Stochastic Contextual Bandits / Part 3

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Target tracking

A different measure of performance

Content based on Brégère, Gaillard, Goude, Stoltz [2019]

Target tracking

Known: sG parameter σ^2 ; transfer function $\varphi : \mathbb{R}^m \times [K] \rightarrow \mathbb{R}^d$

Unknown: reward parameter θ^*

At each round $t = 1, 2, \dots$,

0. A context $\mathbf{x}_t \in \mathbb{R}^m$ and a target $\mathbf{c}_t$ are revealed

No assumption on their generation

1. Statistician picks $A_t \in [K]$

2. She gets a reward $Y_t = \varphi(\mathbf{x}_t, A_t)^\top \theta^* + \eta_t$

where η_t is an independent σ^2 -sG noise

3. This reward Y_t is the only feedback she receives

Performance measured through $\text{loss} (Y_t - \mathbf{c}_t)^2$

Not reward Y_t anymore

→ Rather mix over the arms

Both for the algorithm and the comparison class

Target tracking, by mixing over arms

Arm-by-arm reward model: $Y_{t,a} = \varphi(\mathbf{x}_t, a)^\top \theta^* + \eta_{t,a}$

Mixing over the arms $\leftarrow$ probability distr. $p_t = (p_{t,a})_{a \in [K]}$

$$Y_{t,p_t} = \underbrace{\sum_{a \in [K]} p_{t,a} \varphi(\mathbf{x}_t, a)^\top \theta^*}_{= \varphi(\mathbf{x}_t, p_t)} + \underbrace{\sum_{a \in [K]} p_{t,a} \eta_{t,a}}_{= p_t^\top \eta_t}$$

Joint noise assumptions $\eta_t = (\eta_{t,a})_{a \in [K]}$ are crucial
vs. $\varphi(\mathbf{x}_t, p_t)$ linear in p_t is unimportant

E.g., vectors η_t are σ^2 -sub-Gaussian (thus centered)
i.i.d. with $\Gamma = \text{Var}(\eta_t)$

Target tracking, by mixing over arms — **what regret?**

$$Y_{t,p_t} = \varphi(\mathbf{x}_t, p_t)^\top \theta^* + p_t^\top \eta_t$$

where η_t is an independent σ^2 -sG noise with $\text{Var}(\eta_t) = \Gamma$

$$\sum_{t=1}^T (Y_{t,p_t} - c_t)^2 \quad \text{should be close to}$$

$$\sum_{t \leq T} \mathbb{E} \left[(Y_{t,p_t} - c_t)^2 \mid \mathcal{F}_{t-1} \right] = \sum_{t \leq T} (\varphi(\mathbf{x}_t, p_t)^\top \theta^* - c_t)^2 + \overbrace{\mathbb{E} \left[(p_t^\top \eta_t)^2 \mid \mathcal{F}_{t-1} \right]} = p_t^\top \Gamma p_t$$

$$\rightarrow \text{Pseudo-regret:} \quad \sum_{t \leq T} (\varphi(\mathbf{x}_t, p_t)^\top \theta^* - c_t)^2 + p_t^\top \Gamma p_t$$

$$\text{vs.} \quad \sum_{t \leq T} \min_p \left\{ (\varphi(\mathbf{x}_t, p)^\top \theta^* - c_t)^2 + p^\top \Gamma p \right\}$$

Target tracking, by mixing over arms

Known: sG parameter σ^2 ; transfer function $\varphi : \mathbb{R}^m \times [K] \rightarrow \mathbb{R}^d$

Unknown: reward parameter θ^* ; noise covariance matrix Γ

At each round $t = 1, 2, \dots$,

0. A context $\mathbf{x}_t \in \mathbb{R}^m$ and a target c_t are revealed
No assumption on their generation
1. Statistician picks probability distribution p_t over $[K]$
2. She gets a reward $Y_{t,p_t} = \varphi(\mathbf{x}_t, p_t)^\top \theta^* + p_t^\top \eta_t$
where η_t is an independent σ^2 -sG noise with $\text{Var}(\eta_t) = \Gamma$
3. This reward Y_{t,p_t} is the only feedback she receives

Goal: Minimize cumulative losses $\longleftrightarrow$ Minimize pseudo-regret

$$R_T = \sum_{t \leq T} (\varphi(\mathbf{x}_t, p_t)^\top \theta^* - c_t)^2 + p_t^\top \Gamma p_t - \sum_{t \leq T} \min_p \left\{ (\varphi(\mathbf{x}_t, p)^\top \theta^* - c_t)^2 + p^\top \Gamma p \right\}$$

LinUCB-type strategy for target tracking

→ **Learn** θ^* (estimate while playing): same bounds as before

$$\hat{\theta}_{t-1} = \left(\lambda \text{Id}_d + \sum_{s \leq t-1} \varphi(\mathbf{x}_s, p_s) \varphi(\mathbf{x}_s, p_s)^\top \right)^{-1} \sum_{s=1}^{t-1} \varphi(\mathbf{x}_s, p_s) Y_{s,p_s}$$

→ **Pure exploration** for Γ over $n_0 = \mathcal{O}(T^{2/3})$ rounds

Use Dirac masses at $a \in [K]$ or 50%–50% distributions over $a \neq a'$

Estimation based also on $\hat{\theta}_{n_0-1}$ / Pure exploration not so satisfactory of course...

→ Confidence intervals $\left([\hat{\theta}_{t-1}^\top \phi(x_t, p)]_{\text{clip}} - c_t \right)^2 + p^\top \hat{\Gamma} p \pm \alpha_{t,p}$

on the conditional errors $(\theta^\top \phi(x_t, p) - c_t)^2 + p^\top \Gamma p$; clipping cf. boundedness ass.

→ Optimism:

$$p_t \in \arg \min_p \left\{ \left([\hat{\theta}_{t-1}^\top \phi(x_t, p)]_{\text{clip}} - c_t \right)^2 + p^\top \hat{\Gamma} p - \alpha_{t,p} \right\}$$

LinUCB-type strategy for target tracking: **regret analysis**

Note: argmin over non-convex func. of p ; mitigation: restrict $\mathcal{P}$ to a grid in practice

$$p_t \in \arg \min_p \left\{ \left([\hat{\theta}_{t-1}^\top \phi(x_t, p)]_{\text{clip}} - c_t \right)^2 + p^\top \hat{\Gamma} p - \alpha_{t,p} \right\}$$

where $\alpha_{t,p} = \underbrace{\varepsilon_{t,p}}_{\text{for } \theta^*} + \underbrace{\gamma_{n_0,p}}_{\text{for } \Gamma}$

W.h.p.:

$$\begin{aligned} R_T &\leq \square n_0 + 2 \overbrace{\sum_{t \leq T} \alpha_{t,p}}^{\text{as before}} \\ &\leq \square n_0 + 2 \underbrace{\sum_{t \leq T} \varepsilon_{t,p_t}}_{\leq \tilde{O}(\sqrt{T})} + 2T \underbrace{\gamma_{n_0,p}}_{\leq \tilde{O}(1/\sqrt{n_0})} \leq \tilde{O}(T^{2/3}) \quad \text{for } n_0 \sim T^{1/3} \end{aligned}$$

Some heavy maths involved in estimating Γ

Application

Managing electricity demand by sending price incentives

Motivation

Aim: maintain balance between production and consumption

Current solution: forecast consumption and adapt production

Prospective solution: encourage/discourage consumption
by dynamically setting prices



Bandit monitoring: trade-off between

Learning behaviors of customers (= exploration)

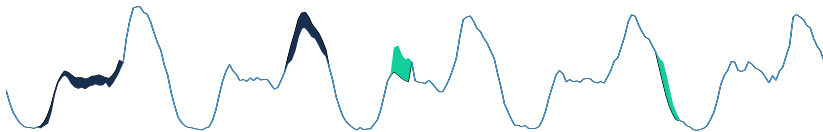
Optimizing incentives sent (= exploitation)

→ Stochastic bandit theory should be applicable...

Motivation: data set

“SmartMeter Energy Consumption Data in London Households”
Public dataset – by UK Power Networks

Individual consumptions at half-hourly frequency in year 2013
About 1,000 customers with tariff incentives



K=3 tariffs: **Low (L)**, **Normal (N)**, **High (H)**

Modeling consumption by a known and effective methodology

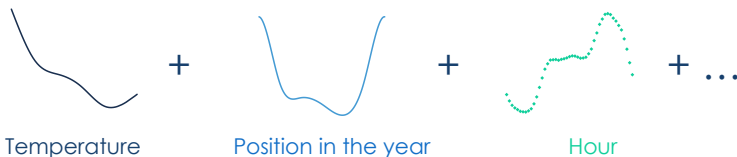
Population assumed to be homogeneous (as a first approach)

Mean consumption Y depends on context $x_t \in \mathbb{R}^d$

Temperature, season, day of the week, hour of the day, etc.

Also depends on tariff $a \in \{1, \dots, K\}$

$$Y_t = f_1(\text{temperature}) + f_2(\text{position in the year}) + f_3(\text{hour}) + (\quad) + \dots + \text{noise}$$



Plain additive effect for tariffs

If single tariff a picked:

$$Y_{t,a} = \mathbf{e}_a + \sum_{i=1}^d f_i(x_{t,i}) + \text{noise}$$

Modeling consumption by a known and effective methodology

Context $x_t \in \mathbb{R}^d$ and tariff a :
$$Y_{t,a} = e_a + \sum_{i=1}^d f_i(x_{t,i}) + \text{noise}$$

Generalized additive model

→ effective modeling

(Wood, 2006; Goude et al., 2014; Gaillard et al., 2016)

- The f_i are cubic splines
- We fix the number q_i of knots and their location
- There exists a basis $b_1^{(i)}, \dots, b_{q_i}^{(i)}$
- We write $f_i = \sum_{1 \leq j \leq q_i} \beta_j^{(i)} b_j^{(i)}$ for each i

Summary: for x_t and a ,

$$Y_{t,a} = e_a + \beta^\top \phi(x_t) + \text{noise}$$

where β and e_a are unknown, but $\phi(x_t)$ is known

Modeling consumption → distributions of tariffs

$$Y_{t,a} = e_a + \beta^\top \phi(x_t) + \text{noise}$$

Tariffs $\{1, \dots, K\}$ distributed in shares $p_t = (p_{t,1}, \dots, p_{t,K})$:

$$\begin{aligned} Y_{t,p_t} &= \sum_{a \in [K]} p_{t,a} Y_{t,a} = \sum_{a \in [K]} p_{t,a} (e_a + \beta^\top \phi(x_t) + \eta_{t,a}) \\ &= \varphi(x_t, p_t)^\top \theta^* + p_t^\top \eta_t \end{aligned}$$

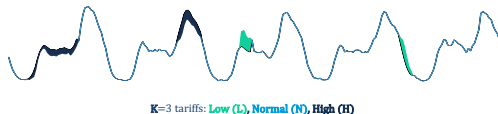
with θ^* unknown, but $\varphi(x_t, p_t)$ is known (and linear in p_t)

Noise vectors: η_t i.i.d., σ^2 -sub-Gaussian, $\text{Var}(\varepsilon_t) = \Gamma$ unknown

We use this model as a data generator to test our bandit strategies
(cf. impossible on historical data!)

Realistic simulations

Reminder: we have a historical data set with $K = 3$ tariffs and p_t chosen as Dirac masses



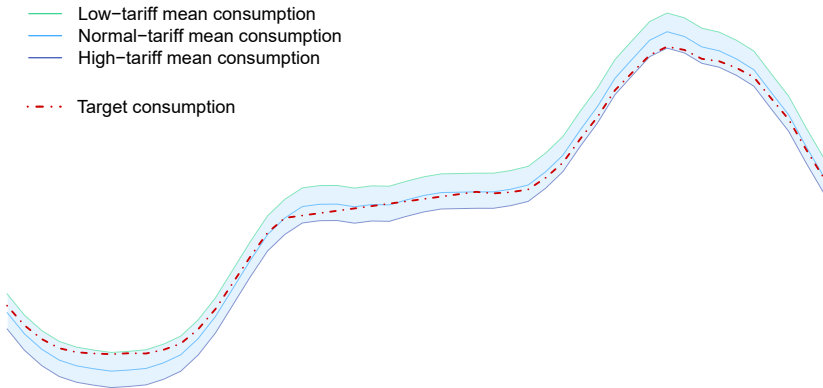
We are bound to generate new data

We estimate our GAM model with 1 year of such data

We then generate new consumption data
based on this model + historical weather variables

We create attainable targets: $\phi(x_t, \text{high})^\top \theta^* \leq c_t \leq \phi(x_t, \text{low})^\top \theta^*$

Realistic simulations



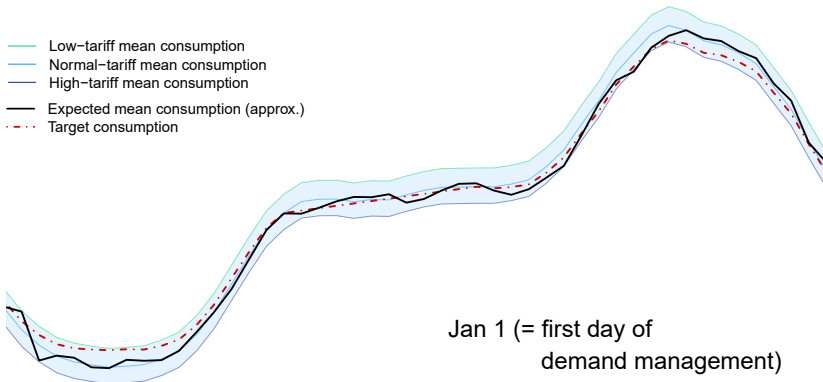
Aim: smooth out consumption

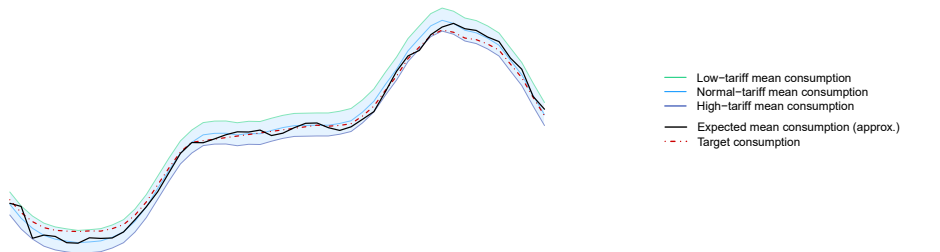
Experiment design

→ provider changing its policy

- Pick the “normal” tariff for 1 year, i.e., $p_t = (0, 1, 0)$
- Then start picking different allocations with at most 2 tariffs (either [low+normal] or [normal+high])

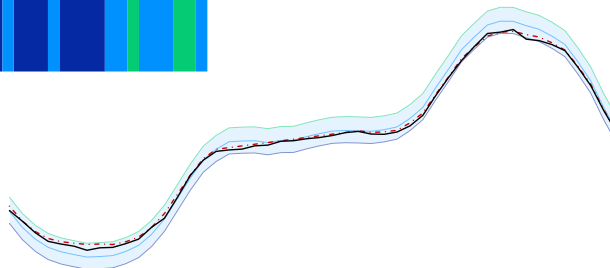
Repeat this 200 times





Top: January 1

Bottom tariff allocations
based on a single run



Bottom: January 30

What's next?

1. Meta-learning of flexibility

Clusters of clients with unknown and different flexibility levels

2. Rebound effect

→ Axis 1.2 of the EDF–Inria project

Post-docs of Julien Aubert + Eliabelle Mauduit